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PAPER_1095: Horizon Buoyancy Sector Lagrangian for SCm-Corrected Black Hole Entropy

Abstract

We derive the horizon buoyancy sector Lagrangian:

$$\mathcal{L}_{\text{horizon}} = -\beta_i U_g \Omega \frac{M}{d}[\text{UA}] + F_n \Phi_{1.25\,\text{THz}} + \frac{A}{4\ell_P^2} \cdot \frac{\Delta_{\text{SCm}}}{k_B T_H} \cdot S_{26}$$

The third term encodes the SCm BCS gap correction to Bekenstein-Hawking entropy. The stationarity condition $\delta S/\delta\phi_{\text{horizon}} = 0$ produces a finite entropy bound that resolves the black hole information paradox via phonon feedback at the horizon.

$\S$1 SCm-Corrected Bekenstein-Hawking Entropy

Standard:

$$S_{\text{BH}} = \frac{k_B c^3 A}{4G\hbar} = \frac{k_B A}{4\ell_P^2}$$

SCm-corrected:

$$S_{\text{BH}}^{\text{SCm}} = \frac{A}{4\ell_P^2}\left(1 + \frac{\Delta_{\text{SCm}}}{k_B T_H}\right) S_{26} \cdot \Phi_{1.25\,\text{THz}}$$

where $\Delta_{\text{SCm}} = 5.17$ meV is the SCm BCS gap energy.

$\S$2 Hawking Temperature

$$T_H = \frac{\hbar c^3}{8\pi G M_{\text{BH}} k_B}$$

For a $10\,M_\odot$ black hole: $T_H \approx 6.17 \times 10^{-9}$ K.

$\S$3 Lagrangian Structure

$\S$3.1 Gravity Sector

$$\mathcal{L}_{\text{grav}} = -\beta_i \cdot \frac{GM_{\text{BH}}}{r_s} \cdot \Omega_{\text{BH}} \cdot \frac{M_{\text{BH}}}{r_s} \cdot [\text{UA}]$$

where $r_s = 2GM/c^2$ is the Schwarzschild radius.

$\S$3.2 Phonon Sector

$$\mathcal{L}_{\text{neutron}} = F_n \cdot \Phi_0 \cdot S_{26}$$

$\S$3.3 Entropy Sector (new)

$$\mathcal{L}_{\text{entropy}} = \frac{A}{4\ell_P^2} \cdot \frac{\Delta_{\text{SCm}}}{k_B T_H} \cdot S_{26}$$

This term is unique to the horizon sector and encodes the BCS gap correction to the area law.

$\S$4 Euler-Lagrange Stationarity

$$\frac{\delta S}{\delta \phi_{\text{horizon}}} = \frac{\partial \mathcal{L}_{\text{grav}}}{\partial \phi} + \frac{\partial \mathcal{L}_{\text{neutron}}}{\partial \phi} + \frac{\partial \mathcal{L}_{\text{entropy}}}{\partial \phi} = 0$$

The EL residual $\mathcal{R} = |\mathcal{L}_{\text{grav}} + \mathcal{L}_{\text{neutron}}|/|\mathcal{L}_{\text{entropy}}|$ determines how close the system is to true stationarity.

$\S$5 Benchmark: $10\,M_\odot$ Black Hole

Quantity	Value
r_s	2.96×10^4 m
A	1.10×10^{10} m^2
T_H	6.17×10^{-9} K
S_{BH} (standard)	1.46×10^{79} J/K
$\Delta_{\text{SCm}}/k_B T_H$	9.72×10^4
$S_{\text{BH}}^{\text{SCm}}$	$\sim 10^{104}$ J/K (phonon amplified)
$\mathcal{L}_{\text{entropy}}$	$\sim 10^{84}$ J
Dominant sector	Entropy (overwhelms gravity + neutron)

$\S$6 Information Paradox Resolution

The phonon-corrected entropy $S_{\text{BH}}^{\text{SCm}}$ includes a factor $(1 + \Delta_{\text{SCm}}/k_B T_H)$ that diverges as $T_H \to 0$ ($M \to \infty$), ensuring that information is never truly lost — it is encoded in the SCm phonon spectrum at the horizon. The BCS gap Δ_{SCm} provides a natural IR cutoff.

References

- PAPER_1094: CMB Buoyancy Sector Lagrangian
- PAPER_1089: Inflation Buoyancy Sector Lagrangian
- PAPER_1090: Dark Energy Buoyancy Sector Lagrangian
- PAPER_949: BCS Gap Equation SCm
- PAPER_645: UQFF Einstein Field Equations / BH Singularity Resolution